Unit 3 Progress Check: MCQ Part A

1. A student is developing a class that will store information about pets, including each pet's name and age. Which of the following sets of attributes is most appropriate for this class?

(A) One String attribute that includes the pet's name and age

(B) One String attribute for the pet's name and one String attribute for the pet's age

(C) One String attribute for the pet's name and one int attribute for the pet's age ✓

(D) One int attribute for the pet's name and one int attribute for the pet's age

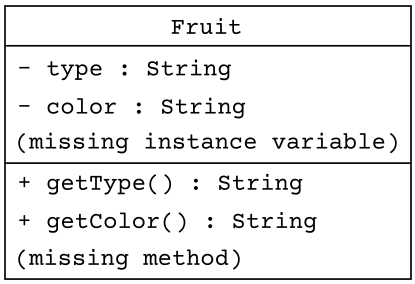
Answer C

Correct. Names are best represented using string variables. Ages typically refer to whole numbers, so they are best represented using integer variables.

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2. The Fruit class contains attributes for a fruit's type, color, and quantity. The class also contains methods to allow the attributes to be accessed outside the class.

A class design diagram for the Fruit class contains three sections. The first section contains the class name, the second section contains three instance variables and their data types, and the third section contains three methods and their return types. The + symbol indicates a public designation, and the - symbol indicates a private designation.



Which of the following are the most appropriate replacements for (missing instance variable) and (missing method)?

(A) Replacing (missing instance variable) with - quantity : String and replacing (missing method) with + getQuantity() : String

(B) Replacing (missing instance variable) with - quantity : int and replacing (missing method) with + getQuantity() : int ✓

(C) Replacing (missing instance variable) with - quantity : int and replacing (missing method) with + getQuantity() : String

(D) Replacing (missing instance variable) with - quantity : String and replacing (missing method) with + getQuantity() : int

Answer B

Correct. Quantities of fruits are nonnegative integers, so the quantity instance variable and the return type of the getQuantity method is best represented using the data type int.

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3. A student is developing a Player class to represent a player in a game. The class should allow other classes to access and modify the player's score.

Which of the following sets of attributes and behaviors is most appropriate for the Player class design?

(A)

Attributes:	none
Behaviors:	setScore and getScore

(B)

Attributes:	setScore and getScore
Behaviors:	none

(C)

Attributes:	getScore and setScore
Behavior:	score

(D) ✓

Attribute:	score
Behaviors:	getScore and setScore

Answer D

Correct. The class needs to maintain a score, which can be done using an attribute. The class needs to allow other classes to view and modify the score, which can be done using behaviors.

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4. A programmer wants to speed up the development of a program by reducing the amount of testing performed on the program. Which of the following is most likely to be a consequence of this approach?

(A) Decreased intellectual property concerns

(B) Decreased system reliability ✓

(C) Increased intellectual property concerns

(D) Increased system reliability

Answer B

Correct. Programmers should test the program with a variety of conditions to maximize system reliability. Reducing the amount of testing may negatively affect system reliability.

5. A software company created a program to help solve a problem in a community. Which of the following is a true statement about the program?

(A) The program can be reused by other programmers without obtaining permission.

(B) The program could have unintended harmful effects beyond its intended use. ✓

(C) The program should be implemented with as few abstractions as possible.

(D) The program will improve the system reliability of other programs.

Answer B

Correct. Not every effect of a program is anticipated in advance. A program designed to solve a problem may also have a harmful impact on society, the economy, or culture.

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6. Which of the following best explains why programmers often include open source code in their programs?

(A) To utilize algorithms that eliminate the need to include comments in program code

(B) To utilize algorithms that have already been implemented while reducing the risk of raising intellectual property concerns ✓

(C) To utilize algorithms that minimize the need for abstraction in a program

(D) To utilize algorithms that remove the requirement to design a program before implementing it

Answer B

Correct. Open source licenses allow programmers to reuse code written by others. These licenses often allow code to be used freely without the need for special legal permissions.

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7. Consider the following partial class declaration.

```
public class Item
{
  private double price; // The price of the item
  public String name;   // The name of the item

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The following code segments each appear in a class other than Item. Assume that myItem and myItem2 are Item objects that have been properly instantiated.

Code Segment I

```
int x = myItem.price;
System.out.println(x);
```

Code Segment II

```
String y = myItem2.name;
System.out.println(y);
```

What, if anything, is printed as a result of executing each of the code segments?

(A) Code segment I prints the price of myItem, and code segment II prints the name of myItem2.

(B) Code segment I prints the price of myItem, but code segment II does not compile.

(C) Code segment II prints the name of myItem2, but code segment I does not compile. ✓

(D) Nothing is printed because neither code segment compiles.

Answer C

Correct. The instance variable price is declared as private, so it cannot be accessed outside the class. This will prevent code segment I from compiling. The instance variable name is declared as public, so it can be accessed outside the class. This allows the name to be printed in code segment II.

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8. The following Candy class is used to represent pieces of candy.

```
public class Candy          // line 1
{
  public String name;       // line 3
  public double weight;     // line 4

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The class is intended to allow external classes to create Candy objects, but external classes should not be able to modify the attributes of Candy objects. Access to the class is not constrained as intended.

Which of the following changes can be made so that access to the class is constrained as intended?

(A) Changing the public designation to private in line 1

(B) Changing the class header in line 1 to public class Candy()

(C) Changing the public designation to private in lines 3 and 4 ✓

(D) Changing the data type in line 4 from double to String

Answer C

Correct. In order to prevent external classes from directly accessing and modifying instance variables, they should be declared with the keyword private.

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9. The Profile class will be used to represent users of a social media website. Each instance of the class will maintain the user's name, username, and number of followers. This information is unique to each user and should not be directly accessible outside the Profile class.

Which of the following declarations is the most appropriate for the Profile class?

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(A)

```
private class Profile
{
  private String name;
  private String username;
  private int followers;
}
```

(B)

```
public class Profile
{
  public String name;
  public String username;
  public int followers;
}
```

(C) ✓

```
public class Profile
{
  private String name;
  private String username;
  private int followers;
}
```

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(D)

```
private class Profile
{
  public String name;
  public String username;
  public int followers;
}
```

Answer C

Correct. This properly declares a `public` class and three `private` instance variables to store the attributes of a `Profile` object.

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10. Consider the following class definition.

```
public class Person
{
  private String name;
  /* missing constructor */
}
```

The following statement, which is located in a method in a different class, creates a new Person object with its attribute name initialized to "Washington".

```
Person p = new Person("Washington");
```

Which of the following can be used to replace /* missing constructor */ so that the object p is correctly created?

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(A)

```
private Person(String n)
{
  name = n;
}
```

(B)

```
public Person()
{
  name = n;
}
```

(C) ✓

```
public Person(String n)
{
  name = n;
}
```

(D)

```
public Person(String name)
{
  String n = name;
}
```

Answer C

Correct. The constructor is designated as public so that it can be accessed outside the class. The constructor takes

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the String parameter n and assigns it to the instance variable name.

11. Consider the following Student class.

```
public class Student
{
  private String name;
  private int age;

  public Student(String studentName, int studentAge)
  {
    name = studentName;
    studentAge = age;
  }
}
```

The following code segment appears in a class other than Student.

```
Student myStudent = new Student("Bobby", 25);
```

Which of the following best describes the state of the object myStudent as a result of executing this code segment?

(A) myStudent has been initialized such that name contains the string "Bobby" and age is 0. ✓

(B) myStudent has been initialized such that name contains the string "Bobby" and age is 25.

(C) myStudent has been initialized such that name is null and age is 0.

(D) myStudent has been initialized such that name is null and age is 25.

Answer A

Correct. The name instance variable is assigned the value of studentName, which in this case is a reference to the String "Bobby". However, the age instance variable is not assigned a value in the constructor, so it holds the default value for an int variable, which is 0.

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12. Consider the following class definition.

```
public class Dog
{
   private String name;
   private int age;

   public String getName()
   {
      return name;
   }

   public int getAge()
   {
      return age;
   }
}
```

The following segment appears in a class other than Dog.

```
Dog fido = new Dog();
System.out.print("name: " + fido.getName());
System.out.print(", age: " + fido.getAge());
```

What is printed as a result of executing this code segment?

(A)

name: , age: 0

(B) ✓

name: null, age: 0

(C)

name: null, age: null

(D) Nothing is printed. An error occurs because the Dog class does not contain a no-parameter constructor.

Answer B

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Correct. Since no constructor is written, Java provides a no-parameter constructor, which sets the String instance variable name to its default value of null and the int instance variable age to its default value of 0.